

Derrick Chen

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EDUCATION

Georgia Institute of Technology

Expected May 2028

Bachelor of Science in Computer Science

- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Object Oriented Programming, Computer Architecture, Machine Learning, Operating Systems, Database Design

EXPERIENCE

Software Development Engineer Intern

Aug 2026 - Present

IBM

San Jose, CA

- Reduced LLM inference costs by designing a turn-level routing engine that dynamically selects between models based on prompt complexity, agent layer requirements, and tool executions
- Decreased agent latency by routing low complexity requests to faster models while preserving automatic escalation to stronger models for complex tool driven workflows
- Maintained a **95%+ quality pass rate** while reducing high cost model usage through heuristic complexity classification, judge-based evaluation, and retry-aware model escalation

Software Engineer Intern

June 2026 - Aug 2026

Walmart

Bentonville, AR

- Designed an event-driven stocking confirmation flow processing **3,000+ requests/minute**, replacing inventory guesswork with real-time stocking updates saving a store hundreds of thousands annually
- Implemented the backend with Java, Spring Boot, and Apache Kafka, implementing event mapping, validation, and publishing logic with **90%+ code coverage**
- Built a Grafana and Prometheus/Apache JMeter dashboard tracking **p95/p99 latency**, HTTP throughput, Kafka publish rates, and CPU/heap utilization to monitor production metrics

Software Engineer Intern

June 2025 - Aug 2025

BigID

Atlanta, GA

- Developed an integration for **BigID customers** to stream DSPM cases into AWS Security Lake using **open source OCSF standard**, allowing customers to centralize DSPM data within their AWS environments
- Built the backend in Node.js, using ETL pipelines to handle **1+ TB** of security data, cutting ingestion time by over **100x (hours to minutes)**

Undergraduate Researcher

Aug 2024 - Dec 2024

GROWER Research Lab

Atlanta, GA

- Owned the creation of a **Python** data analytics pipeline that scraped power outage data from **13+ sources** and cleaned **1M+ records**, expanding dataset coverage by **15%** of U.S. states

PROJECTS

Referee Ruling Engine | *Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector*

ask-the-ref.com

- Built an NBA rules verification engine averaging **1.6K users/visits monthly** that translates **natural language** basketball scenarios into a legal or illegal verdict
- Designed a hybrid **RAG retrieval system** combining **pgvector similarity search** and **PostgreSQL full-text search** to surface the most relevant rule sections for each scenario
- Engineered a structured reasoning pipeline that extracts scenario facts such as **timing, possession, location, and intent**, then evaluates them exclusively against retrieved official rules to reduce unsupported model judgments

LeetCode Auto Clear | *JavaScript, Chrome Extensions API*

[Chrome Web Store](#)

- Built a Chrome extension with **590+ users** that automatically clears previous solutions to prevent them from spoiling future practice

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C++

Frameworks: FastAPI, React, Node.js, Next.js, Flask, Django, Spring Boot, Express

Developer Tools: AWS, Git/GitHub, Vercel, Jira, Playwright, Azure, PyTest, Docker, Kubernetes, Kafka

Databases: PostgreSQL, MongoDB, MySQL

AI Development: Model Context Protocol (MCP), AI Agents, Claude Code, Cursor, Prompt Engineering

EXTRA

Awards: Coca-Cola First Generation Scholar, Zell Miller Scholarship Recipient

Extracurriculars: WebDev @ GT, Intramural Sports, Blockchain @ GT, Greek Life (Director of Career Development)